

National School Lunch Program Certification Error Risk Assessment Tool



Prepared by Mathematica Policy Research

The Child Nutrition and WIC Reauthorization Act of 2004 requires additional administrative reviews with local education agencies that have demonstrated a high level of, or a high risk for, certification error.

Acronyms

AAR = additional
administrative review

F/RP = free or reduced-price

LEA = local education agency

SA = State agency

SFA = school food authority

VSR = Verification Summary
Report

What is Certification Error Risk?

Certification errors occur when students are incorrectly certified for NSLP free or reduced-priced (F/RP) meals for which they are not eligible, or incorrectly denied benefits for which they are eligible.

The *Access, Participation, Eligibility and Certification (APEC) Study* found that one in five children certified for F/RP meals in SY2005-2006 was erroneously certified or incorrectly denied benefits, resulting in erroneous payments of \$759 million nationwide (9% of NSLP reimbursements). Most errors result from overpayments due to household misreporting, with only a small portion due to administrative error.

Certification error risk is a measure of the likely dollar value of NSLP meal benefits provided to students whose certification status is in error.

This pamphlet describes:

- Statistical modeling to identify indicators of certification error risk,
- A risk assessment tool to identify SFAs at high risk.

The risk assessment tool is designed for State Child Nutrition Agencies. The tool identifies high risk SFAs that are candidates for additional administrative reviews (AARs).

Purpose of the Tool

The NSLP risk assessment tool can be used by State Agencies (SAs) to select SFAs for AARs.

According to USDA Food and Nutrition Service (FNS) guidance (SP-48-2011), SAs must annually conduct AARs of selected SFAs that meet any of the following criteria:

1. Establishment of a new Provision 2/3 base year since the previous CRE was conducted.
2. Verification data indicating a high level (e.g., top 25% among SFAs within a State) of non-response or response-based terminations.
3. Consistently claiming over 90% free eligible or 80% reduced-price eligible.

The risk assessment tool provides SFA information relative to these criteria in an easy-to-use format. The tool also identifies SFAs with high risk of certification error based on statistical modeling.

SAs are required to meet FNS requirements for a minimum number of AARs:

1. **1% of SFAs:** SAs must conduct AARs on a number of SFAs equivalent to no less than 1% of the SFAs in the State; and
2. **3% of total annual reimbursement:** SAs must conduct AARs on a number of SFAs comprising no less than 3% of the total statewide reimbursement.

The risk assessment tool measures AAR selections against these two size criteria.

Development of the Risk Assessment Tool

Step 1: Statistical Modeling

It is not possible to measure accurately the certification error rates of each SFA each year (see sidebar). However, if we know the relationship between SFA characteristics and error rates, we can obtain **estimates** of error rates.

Based on statistical models, we know that different SFA characteristics affect different types of certification error. Six statistical models were developed using data from the APEC study, which measured error rates for a national sample of SFAs. The six models show the relationship between SFA characteristics and **six types of error rates** (see below).

The most important SFA characteristics related to error rates include:

- SFA verification results
- SFA enrollment
- Whether an SFA is public or private
- Percentage of students in each NSLP certification category: free, reduced-price, paid; not subject to verification; Provision 2/3

NSLP verification results provide indicators of certification risk, but do not provide precise and unbiased measures on their own because:

- (a) verification samples are sometimes very small, and*
- (b) most verification samples are not random.*

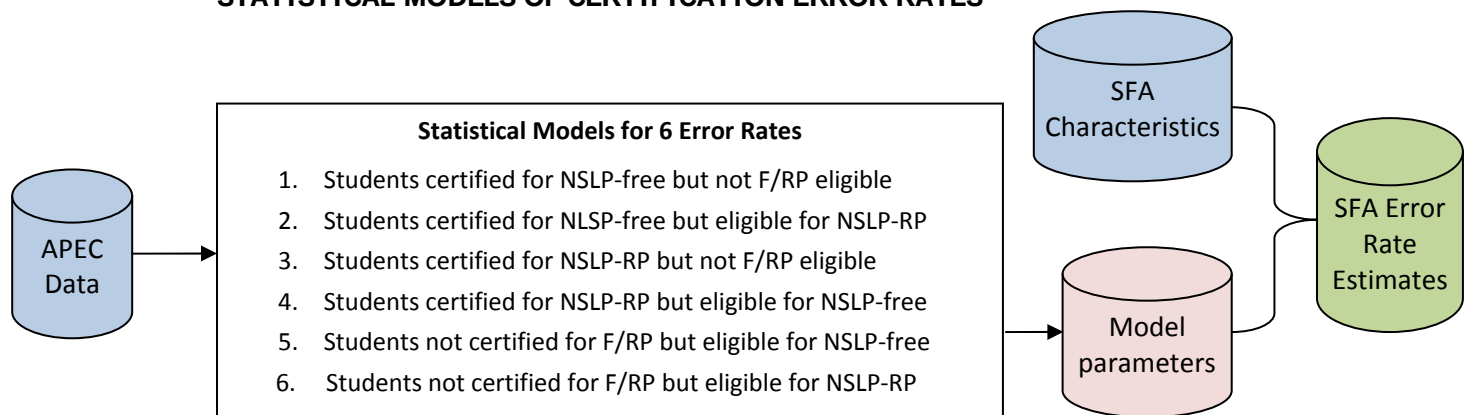
Step 2: Using Model Results in a Web-Based Tool

The risk assessment tool combines the model estimates of error rate relationships with SFA characteristics to calculate the six estimated error rates. SFAs have higher error rate estimates if their characteristics are similar to SFAs from the APEC study that had high error rates. Annual measures of SFA characteristics are taken from the VSRs submitted by SFAs.

The total error rate is calculated for each SFA from the six separate estimates, weighted by the SFA's distribution of free and reduced-price applications. This single estimate of total error is multiplied by SFA reimbursements to provide an estimate of the dollar value of certification error. The SFA dollar values of certification error are used to rank SFAs in a State by certification error risk.

For technical details, see the full study report, referenced on the back page.

STATISTICAL MODELS OF CERTIFICATION ERROR RATES



What is High Risk?

Risk is determined by the estimated dollar value of certification error. SFA risk depends on two factors:

1. Estimated certification error rate for the SFA
2. Total annual NSLP dollar reimbursements for the SFA

Very small SFAs with high error rates have low risk (few dollars are at risk).

Risk Categories

The risk assessment tool includes a risk score and risk category for each SFA. The risk score is a value from 0 to 100. The risk category is high, medium, or low.

Risk score is based on estimated dollars reimbursed in error. This value is converted to a scale of 0 to 100 so that relative risk can be compared across

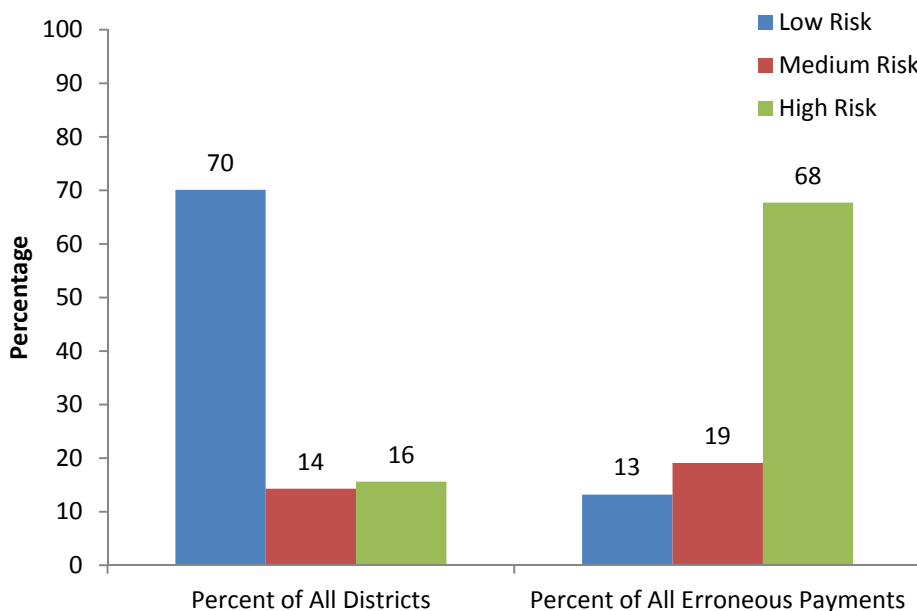
States and over time. All high risk SFAs have a risk score of 50 or higher.

Two adjustments are made when assigning SFAs to the high, medium, or low risk categories.

- The thresholds for high, medium, and low risk vary by State to account for differences across States in average SFA enrollment (see definition of risk categories on back page).
- SFAs with estimated error rates of five percent or less are not categorized as high risk, regardless of the dollar value of error. This is because it may be very difficult to reduce error to less than 5 percent.

Risk depends on estimated error rates and dollar reimbursements.

DISTRIBUTION OF SFAS AND ERRONEOUS PAYMENT BY RISK CATEGORY

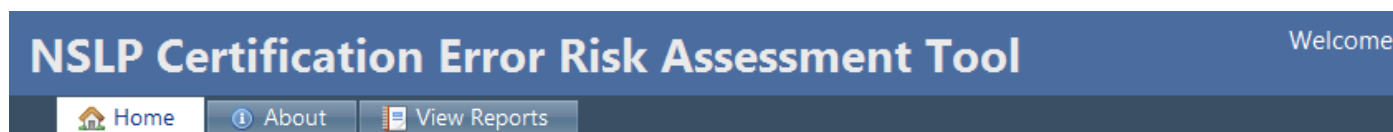


In SY 2009-2010, high risk SFAs accounted for:

- 16% of all SFAs
- 68% of all erroneous payments

Note: Based on certification error risk definition using a five-percent rate floor.

Introducing the Risk Assessment Tool



Home

NSLP Tool for Assessing Certification Error Risk

What is certification error risk? Certification error occurs when students are incorrectly certified for NSLP free or reduced-priced meals for which they are not eligible, or are incorrectly denied benefits for which they are eligible. The APEC study found that one in five children certified for F/RP meals was erroneously certified or incorrectly denied benefits, resulting in erroneous payments.¹ Most errors are due to overpayments and most overpayments are due to household misreporting, with only a small portion due to administrative error.

Certification error risk is the potential dollar value of NSLP meal benefits provided to students whose certification status is in error. For each LEA, this risk depends on the estimated percentage of school meal reimbursements

The tool is available through the USDA PartnerWeb at:

www.partnerweb.usda.gov

Three types of users have access to different levels of information and different functions in the risk assessment tool.

State Agencies

- o View reports with information for SFAs in your State
- o Select SFAs relative to AAR selection criteria
- o Download reports for selected SFAs
- o Download an Excel file with all SFAs

FNS Region Offices

- o Same functions as State users, for all States in the region
- o View administrative reports

FNS Headquarters

- o Same functions as State users, for all States
- o Upload data (annually) and view admin reports
- o Run the model
- o Generate reports

The tool contains five tabs. Three tabs are available to State users.

Home provides information about certification error risk and the purpose of the tool.

About describes tool development. Technical information about model specifications and estimated model parameters is available through links on this tab.

View Reports shows the estimated risk category, risk score, and risk ranking of SFAs in a State. Reports also include some information from the VSR. SAs can use the reports to select SFAs for AARs.

Generate Reports is available only to FNS headquarters and is documented elsewhere.

Admin Reports is available to FNS headquarters and regions to review information about data loaded in the tool.

View Reports

NSLP Certification Error Risk Assessment Tool

Welcome test! [Log Out](#)

[Home](#)
[About](#)
View Reports

View Reports

School Food Authorities [Help](#)

State: Alabama

School Year: SY2009-2010

Risk Category: All

School Type: All

SFA Name:

☐ Show Additional Statistics

[View Details of Selected SFAs](#)

[View Map of Selected SFAs](#)

[Export All SFAs to Excel](#)

[Import Actual SFA Reimbursements](#)

Selected SFAs (0)

% of all SFAs = 0%

% of NSLP meal payments = 0%

NSLP reimbursement = \$0

Default Sort Order

1 2 3 4 5 6 7 8 9 10 ...

Page size: 10

139 SFAs in 14 pages

	Rank of risk score ▲	SFAID	SFA Name	County	Risk Category	Risk Score	Estimated Error Rate	NSLP Meal Payments		Any school in Prov 2/3 base yr in 2009-2010?	Verification Sample			
								Estimated Dollar Amount	% of State total		Type of verification	#Apps sampled	%Did not respond	%Terminated
☒	1	1	District 1	County 1	High	99.96	17.31	\$3,297,622	8%	☒	Alt-random	104	41%	50%
☐	2	2	District 2	County 2	High	99.87	34.55	\$3,063,524	8%	☐	Alt-random	110	30%	43%
☐	3	131	District 131	County 131	High	99.39	51.61	\$2,491,286	6%	☐	Alt-random	31	13%	32%
☐	4	91	District 91	County 91	High	99.38	55.00	\$1,389,208	3%	☐	Alt-random	20	5%	25%
☐	5	411	District 411	County 411	High	99.29	26.53	\$1,341,820	3%	☐	Alt-random	49	24%	24%
☐	6	132	District 132	County 132	High	99.26	27.45	\$1,537,138	4%	☐	Alt-random	51	53%	59%
☐	7	139	District 139	County 139	High	98.97	22.03	\$1,467,717	4%	☐	Alt-random	59	31%	32%
☐	8	93	District 93	County 93	High	98.82	36.84	\$1,268,997	3%	☐	Alt-random	38	11%	24%
☐	9	271	District 271	County 271	High	98.80	31.25	\$1,286,391	3%	☐	Alt-random	16	25%	50%
☐	10	151	District 151	County 151	High	96.86	27.59	\$934,173	2%	☐	Alt-random	29	41%	48%

1 2 3 4 5 6 7 8 9 10 ...

Page size: 10

139 SFAs in 14 pages

There are four ways to control the display of information in reports:

1. Filters at the top of the page

- School year – years from SY2008-2009
- Risk Category – All, Low, Medium, or High
- School Type –All, Public, or Private
- SFA Name – type an SFA name to display only that SFA

2. Show Additional Statistics – Check the box to display four additional columns of information about SFAs:

- a. Approved for F/RP, # applications
- b. Approved for F/RP, # of students
- c. Percent of students, approved for Free-NSLP
- d. Percent of students, approved for RP-NSLP

3. Sort the report – Click a column header to sort the report by that column. Click the column header a second time to sort in reverse order. (Remember you can filter and sort!)

4. Select the page size – Select the number of rows to display on a page using the selector at the top or bottom of the page.

Select SFAs for Additional Administrative Reviews

NSLP Certification Error Risk Assessment Tool

Welcome test! [Log Out]

Home

About

View Reports

View Reports

School Food Authorities [Hide](#)

Click [Help](#) and [Hide](#) to control the display of selection criteria.

Use this report to select SFAs for additional administrative reviews (AARs) ([SP-48-2011](#)). The box at right helps you identify selections that comprise 1% of all of the SFAs in the State, covering at least 3% of the statewide reimbursement. As you select SFAs, the percentages in the box will increase; percentages display in red and change to green when you meet the size criteria for selections. States may use any of the following criteria for selecting an SFA:

- High risk for certification error (Risk Category)
- Verification data indicating a high-level (e.g. top 25% among SFAs within a State) of non-response or response-based terminations (**%Non-response and/or %Terminated is red for SFAs meeting these criteria**)
- Consistently claiming over 90% free eligibles or 80% reduced price eligibles - must be identified from your records (not shown in report)
- Establishment of a new Provision 2/3 base year since the previous CRE was conducted - must be identified from your records and indicated by checking box in report

1

2

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...

Page size: 10

139 SFAs in 14 pages

		SFAID	SFA Name	County	Risk Category	Risk Score	Estimated Error Rate	NSLP Meal Payments			Verification Sample			
☐	Rank of risk score ▲							Estimated Dollar Amount	% of State total	Any school in Prov 2/3 base yr in 2009-2010?	Type of verification	#Apps sampled	%Did not respond	%Terminated
☐	1	1	District 1	County 1	High	99.96	17.31	\$3,297,622	8%	☐	Alt-random	104	41%	50%
☒	2	2	District 2	County 2	High	99.87	34.55	\$3,063,524	8%	☒	Alt-random	110	30%	43%
☐	3	131	District 131	County 131	High	99.39	51.61	\$2,491,286	6%	☐	Alt-random	31	13%	32%

Select SFAs

Use the Risk Assessment Tool to identify SFAs for AARs. Click checkboxes in the first column to select an SFA. Use these methods for selection:

- Filter or sort the **Risk Category** to review and select high risk SFAs.
- Sort the columns with **Verification Sample** statistics to review SFAs with verification nonresponse or termination rates in the "top 25%." These statistics are displayed in **red**.
- View additional statistics and sort the columns with Approved Free-NSLP and Approved RP-NSLP to identify SFAs with high eligibility rates.
- Identify SFAs with a new Provision 2/3 base year since the previous CRE. You must identify these SFAs from your records. You can check the box in the monitoring tool to save the information and sort by this column.

Track SFA Selections Relative to Size Criteria

When you select an SFA, the "Selected SFAs" box (shown on page 5) is automatically updated to track SFA selections against the two size criteria specified by FNS.

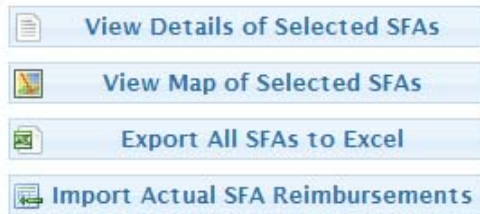
- % of all SFAs = **red** if <1%
= **green** if ≥1%
- % of NSLP meal payments = **red** if <3%
= **green** if ≥3%

The FNS size criteria are satisfied when both measures are displayed in green.

When you uncheck an SFA, the de-selected SFA is removed from the percentages in the SFA Selections box.

Review SFA Selections

Detailed information about SFAs is available using the first three buttons above the report.



 Provides an HTML report with information for each selected SFA. This report contains a link to print to PDF.

 Provides a map with locations of selected SFAs.

 Provides an Excel file with statistics for all SFAs in the State, which you can save to your local computer. Note that selections made in the Excel file will not be available in the online tool for review in future years.

Print SFA details for easy reference during AARs.

Maps can help SAs determine the feasibility of conducting AARs within their travel and time budgets.

Import Actual SFA Reimbursements

One limitation of the tool is that it estimates SFA reimbursements based on total State meal counts. State Child Nutrition Agencies have the option to obtain more accurate risk assessments by importing actual SFA reimbursements for their State. The site provides instructions for this import and a template indicating the required file format.

State:

Year:

Status: **Estimated**

Click the 'Browse / Choose File' button to locate the appropriate reimbursements file and then click IMPORT to load that file into the model.

No file chosen

Note actual SFA amounts will only be uploaded from rows in your file that match the SFA IDs listed for the corresponding year on the reports page. You must include all 3 values (PayPaid, PayRp, and PayFree) in order for the uploaded value to be used.

Here is a [template for the actual SFA Reimbursements file](#) you will need to upload.

States may improve risk score estimates by uploading actual SFA reimbursement data.

Instructions for the upload are available by clicking the button for:

"Import Actual State Reimbursements"

After uploading data, click the link for "[Return to View Reports](#)."

Data Used by the Assessment Tool

The “View Reports” page displays 14 columns of data, expanded to 19 columns when you click “Show Additional Statistics.” Data items and sources are listed below.

Data Items

Col.	Column name	Definition
1	Rank of risk score	Rank order of State SFAs from high to low risk of certification error.
2-3	SFA ID and name	SFA ID and Name (Source: VSR)
4	County	Location of SFA administrative offices (Source: CCD). Displayed if SFA was linked to the Department of Education Common Core of Data (CCD).
5	Risk Category	Districts with risk score above 50 are categorized as high risk, unless they have a very low error rate. All other districts are categorized as medium or low risk. The cutoff between medium and low risk varies by State depending on LEA characteristics.
6	Risk Score	Scores are based on estimated reimbursements made in error. Risk scores range from 0 to 100 because they are normalized to account for differences in average SFA reimbursements across States.
7	Estimated error rate	Number of applications estimated to be approved in error, based on model parameters applied to SFA characteristics
8-9	NSLP Meal Payments: Estimated dollar amount and percent of State total	SFA meal payments are estimates based on: (a) total meal counts for the State, and (b) SFAs’ shares of all NSLP-approved students in the State (calculated separately for free, RP and paid).
10	Any school in Prov. 2/3 base yr?	This column provides checkboxes for States to indicate SFAs with Provision 2/3 base years. Data are saved in the tool.
11	Type of verification	Type of verification sample (Source: VSR)
12	#Apps sampled	Total # applications sampled for verification (Source: VSR)
13	%Did not respond	Percent of applications sampled for verification with benefits terminated due to nonresponse (Source: VSR)
14	%Terminated	Percent of applications sampled for verification, with benefits terminated due to nonresponse or changed to paid (Source: VSR)
Additional Statistics		
15	Approved for F/RP, # applications	# approved free and reduced price applications (Source: VSR)
16	Approved for F/RP, # students	# students on approved free and reduced price applications (Source: VSR)
17	Percent of students, Approved Free-NSLP	Percent of enrolled students approved for Free-NSLP (Source: VSR)
18	Percent of students, Approved RP-NSLP	Percent of enrolled students approved for RP-NSLP (Source: VSR)

Data Sources

Abbreviation	Data Source
CCD	Department of Education, Common Core of Data, Local Education Agency Universe Survey
Meals	SFA ID and Name (Source: VSR)
Rates	Location of SFA administrative offices (Source: Dept. of Education)
Risk Category	Risk categories vary by State and are based on the value of LEA estimated dollars of certification error multiplied by the ratio of State median enrollment to national median enrollment. High risk LEA’s have a value greater than \$50,000; medium risk LEAs have a value greater than \$25,000 and less than \$50,000; low risk LEA’s have a value of \$25,000 or less.

We’re on the Web!

Modeling of High Risk Indicators of Certification Error in the National School Lunch Program technical report can be found at:

www.fns.usda.gov/oane

USDA Guidance on Additional Administrative Reviews

Additional Administrative Reviews and State Retention of Improperly-Paid Funds in School Year 2011-2012 (SP-48-2011), September 6, 2011.

Online at:
<http://www.fns.usda.gov/cnd/Governance/policy.htm>

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